



[ITTALO PEZZOTTI]

CURRICULUM VITAE

2026

I..F.P.E

VIA GIOVANNI CABOTO 4, 20090 SEGRATE, MILAN ITALY

**SENSORIAL AND LEARNING SYSTEM ENGINEERING
INFORMATIC ENGINEER**

+39 3315977719 | ITTALOPEZZOTTI@GMAIL.COM

PERSONAL DETAILS

Surname : Pezzotti Escobar First name: Ittalo Francisco
 Date of birth : 04/09/1973
 Place of residence : Segrate - Milan
 Nationality : Italian - Colombian

EDUCATION & COURSES

Education

Institute	Period	Education & Projects
Postdoc UNAL Clean Energies	15/04/2015 30/04/2017	National University of Colombia, clean energies and photovoltaic cells, and semiconductor films
Postdoc -CNR Medical Devices	01/09/2012 31/03/2015	Biomedical technology Institute bioinstrumentation medical devices applications
Ph.D. – Diploma of secondary level	01/09/2009 25/06/2012	PhD program in Sensorial & learning systems engineering
B.Sc. Diploma of Master-Degree	01/02/1998 28/12/2003	Master-Degree in Informatics Engineering
PS3 – Diploma of post- secondary level	01/07/1992 30/07/1996	Bachelor Degree in Industrial Instrumentation Technology
PS3 – Diploma of secondary level	01/01/1985 25/11/1991	Classic and technical High school diploma

Courses

Period	Description course	Institute	Certificate
05/10/2025	introduction-to-scrum-master-profession	National Learning Service SENA	Attendance certificate
02/10/2025	python Exploratory Data Analysis	National Learning Service SENA	Attendance certificate
30/09/2025	data-science-k-means-clustering-python Foundations	National Learning Service SENA	Attendance certificate
20/03/2024	Addetti ai lavori elettrici: cei 11-27 PES	AISILIA Associazione Italiana salute e sicurezza	Attendance certificate
23/06/2022	Problems inerenti health and safety in working.	National Research Council. (Segrate- Italy)	Attendance certificate
28/11/2017	Electronic laboratory Safety course.	National Research Council. (Segrate- Italy)	Attendance certificate
14/07/2013	Training and information regarding safetycourse.	National Research Council. (Milan- Italy)	Attendance certificate
06/12/2012	Certification of Italian 2B2	university for foreigners of Siena (Roma-Italy)	Attendance certificate
02/02/2012	Design CAD 3D, part design in solid works.	Nuovamacut (Roma Italy)	Attendance certificate
01/01/2012	Fire safety Industrial course.	General Consulting (Roma Italy)	Attendance certificate
09/11/2010	Java development Course.	Archimede Informatics (Roma Italy)	Attendance certificate
15/07/2007	Electrical and technical worker safety course.	Kelly services (Madrid – Spain)	Attendance certificate
12/08/2002	Course of ADSL, G.HDSL, XDSL and routers systems.	Cabletec Ltda (Medellin-Colombia)	Attendance certificate
20/10/1996	PLC programmable logic controller course.	Colombian Polytechnic University(Medellin-Colombia)	Attendance certificate
05/06/1996	Course in Programming language C++.	Colombian Polytechnic University(Medellin-Colombia)	Attendance certificate

WORK EXPERIENCE & ADDITIONAL QUALIFICATIONS

Work experience

Company	Logo	Period	Department and function
Stiima Institute of Intelligent Industrial Technologies		02/2022 - Today	Technical project engineer.
Biomedical technology Institute. Bioinformatics		11/2016 - 12/2019	Technical project engineer.
University national of Colombia (UNAL). Physics school.		04/2015 - 04/2017	Project research engineer.
Biomedical technology Institute (ITB). Bioinstrumentation		09/2012 - 03/2015	Technical project engineer.
Biosensor SRL.		01/2008 - 07/2012	IT project engineer.
Crystallography Institute (IC). Biotechnology		01/2010 - 02/2012	Technical support engineer, SENSBIOSYN project.
Crystallography Institute (IC). Biosensors		01/2009 - 09/2009	Technical support engineer, MULTIBIOPLAT project.
Crystallography Institute (IC). Agrofood		05/2008 - 11/2008	Technical support engineer, NUTRA –SNACKS project.
Eliop (Spanish electricity networks services).		05/2008 - 11/2008	Telecontrol technical engineer.
Ferreteria Tiendas (hardware stores supplies of machines and tools).		04/2004 - 12/2006	Technical service engineer.
EPM (Public utility service company)		04/2002 - 04/ 2003	Telecommunications Technical Engineer.
SIMEX (Integral plastic packaging solutions) Methods and systems		08/1997 - 05/ 2000	Instrumentation and automation technician.
Colombian Polytechnic University J.I.C. instrumentation		11/1996 - 07/1997	Technical service calibration instruments, measurement.

ADDITIONAL QUALIFICATIONS (WORK EXPERIENCE):

EXPERTISE

	Average	Good	Very Good	Excellent
<i>Operating Systems</i>				
Windows (Server/Xp/7/X)	1 Y			
Linux (Ubuntu/Debian/Mint)	1 Y			
iOS	1 Y			
Windows Xp /CE embedded	1 Y			
Android	1 Y			
BeOS	1 Y			
<i>Programming/Software/Design</i>				
C	2 Y			
C++	3 Y			
HTLM	2 Y			
SQL	3 Y			
JAVA	1 Y			
Visual Basic /basic	4 Y			
Fox Pro	1 Y			
Python	6 m			
Pascal	1 Y			
UML free /Visio	1 Y			
Ladder	2 Y			
<i>Equipment/Hardware/Tools</i>				
Eclipse /netbeans	1 Y			
IAR embedded workbench	1 Y			
Arduino/Raspberry/ST/PIC	1 Y			
Keil uVision	1 Y			
Ms Visual Studio	1 Y			
Matlab /Simulink	1 Y			
GIT (Bit bucket) Source Tree	1 Y			
Atmel Studio /Silabs	1 Y			
Labview	1 Y			
Eagle /Spice/Proteus	1 Y			
Origin	1 Y			
Fluorescence Methods	1 Y			
Amperometry systems	1 Y			
cyclic voltammetry	1 Y			
Ride 7	1 Y			

LANGUAGE SKILLS

	SPEAKING	WRITING
Spanish	Mother tongue	Mother tongue
Italian	Very Good	Very Good
English	Good	Good
French	school level	school level

EXECUTIVE SUMMARY

- Hands-on experience in the process of designing, developing and launching a product.
- Self-started, spearheaded various projects and have project management experience.
- Motivated to create smart, futuristic products which serves a cause of improving our life.
- The identification and installation of hydraulic, pneumatic, electrical components, reading/understanding schematics, assembly, test procedures, inspection and repair.
- Assembly of small-scale plant prototypes for the simulation of variables and the tuning of sensors in metrology.
- Experience of electronic, electrical & electro-mechanical repair & maintenance.
- Ability to perform manual operations such as drilling and sawing (for electronic enclosure assembly).
- Experience in the manufacture and assembly of control system hardware, such as control panel & junction box wiring, PCB assembly and cable termination.
- Good understanding of variables such as Flow, Level, Temperature, pressure, current

EXTRACURRICULAR ACTIVITIES

- Works of general mechanics and automotive mechanics such as replacement of spare parts, maintenance and operation of the usury of the parts.
- Attendance at congresses, events and conferences on current topics on technology renewal themes, such as: new components, food, informatics, economics.

HOBBIES

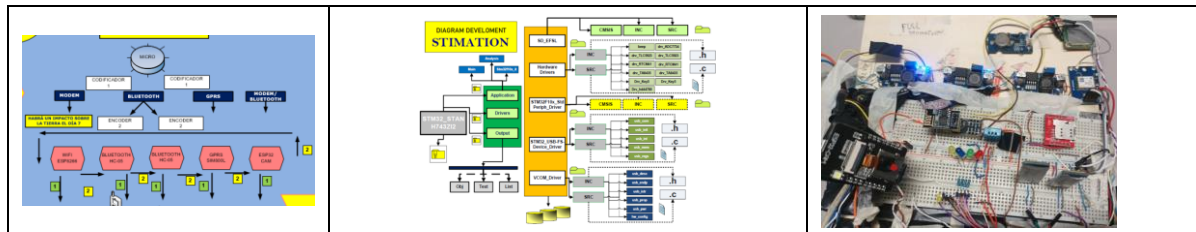
- **Travelling:** Visited 8 + countries. Work or study experience from 3 countries.
- **Robotics:** built several functional robots, for the automation of small processes.
- **Review:** review of all art exhibitions in the control, sensory and instrumental part of the review Sennova, Colciencias y elsevier.
- **Sports:** I perform daily cardiovascular activity, such as: cycling routes, stretching and salon exercises.

PORTFOLIO: STIIMATION COM

DIAGRAM DEVELOPMENT
STIIMATION

Job Title on project | Duration | (01/2024) - (11/2024)

Company	Stiima Institute of Intelligent Industrial Technologies.		
Project Title	Stiimation project consists of the integration of a low-cost communications system for the transmission of encrypted data using different levels of hardware and software security.		
Description Main tasks	• Management of resources and personnel. • Integration of communications devices. • Development of interaction algorithms. • Data flow and information control.		
Sector	-IOT communications / data transmission. -Smart cities traditional networks and services IOT	Areas	• Systems engineering. • Support engineering. • Electronics & Embedded Software.



Prototypes:

- Results:**
- STIIMATION STANDARD: Integration of a low-cost communications system for the transmission of coded data for drones, robots, submarines, and agricultural systems development of an iot station for testing and calibration of gas sensors.
 - STIIMATION FULL: Integration of a low-cost communications system for the transmission of coded data for machines and tools based on network systems.
 - BANK BATTERY SUPERVISOR: Stand-alone battery system for supervised communications systems.

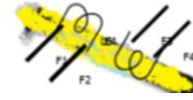
- Responsibilities:**
- Definition of conceptual idea, development and realization
 - Physical and electromechanical assembly of the project prototype.
 - Testing and validation of protocols.
- Activity**
- (Documentation): Drafting of technical documents for reporting.
 - (Documentation): Design and concept (prototype) / industrial design.
 - (Electronics): Test on models and working prototype by computer simulation.
 - (Electronics): Control of hardware/software performance, integrated circuits.
 - (Electronics): Execution of PCB design tests and analysis report drafting.
 - (Mechanics): Creation of individual components and assembly.
 - (Mechanics): (structural dimensioning, kinematics, welding).
 - (Verification): Development of individualized test plans for each project.
 - (Validation): Definition and tracking of basic prototype sensor parameters.



PORTFOLIO: NEWSENTEC

Job Title on project | Duration | (01/2023) - (11/2024)

NEWSENTEC



Company

Stiima Institute of Intelligent Industrial Technologies.

Project Title

NEWSENTEC: New low-cost gas sensors based on tin, titanium and zinc oxides

Description Main tasks

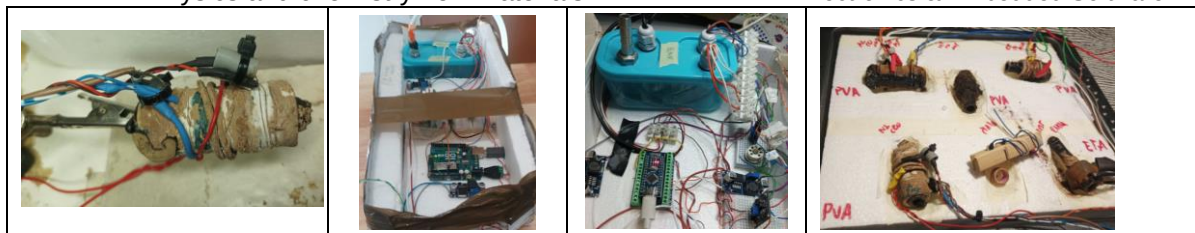
- Development of new gas sensors.
- Development of environmental monitoring systems.
- Development of environmental pollution instruments.

Sector

-Health and medical sciences.
-Work and industry environmental quality.
-Physics and chemistry new materials

Areas

- Systems engineering.
- Support engineering.
- Electronics & Embedded Software.



Projects:

- Gas sensors in the form of conductive arrays.
- Semi-organic gas sensors for environmental contaminants.
- Independent cell with high sensitivity gas sensor for hydrogen.
- Independent cell with gas sensor for inorganic toxic gases.

Results:

Prototypes:

- UCOMICASEG development of an iot station for testing and calibration of gas sensors.
- SENSRESY: Resistance change gas detector for hydrocarbons.
- PROSENTEC: carbon monoxide and carbon dioxide calibration and detection system Resistance change gas detector for hydrocarbons.

Responsibilities:

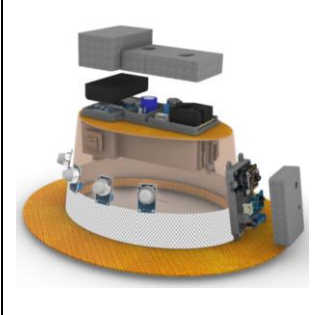
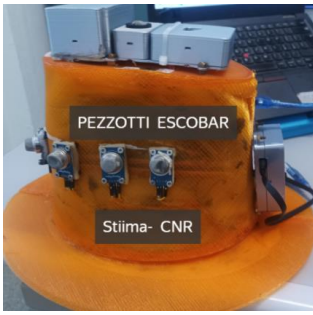
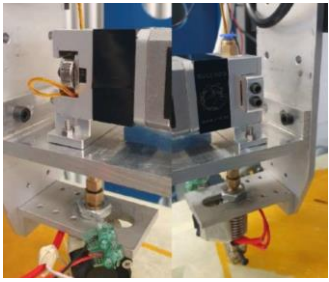
- Management of resources and personnel.
- Definition of conceptual idea, development and realization
- Physical and electromechanical assembly of the project prototype.
- Programming the firmware code.

Activity

- (Documentation): Drafting of technical documents for reporting.
- (Documentation): Design and concept (prototype) / industrial design.
- (Electronics): Test on models and working prototype by computer simulation.
- (Electronics): Control of hardware/software performance, integrated circuits.
- (Electronics): Execution of PCB design tests and analysis report drafting.
- (Mechanics): Creation of individual components and assembly.
- (Mechanics): (structural dimensioning, kinematics, welding).
- (Verification): Development of individualized test plans for each project.
- (Validation): Definition and tracking of basic prototype sensor parameters.

PORTFOLIO: HASNECOD

Job Title on project | Duration | (01/2022) - (11/2024)

Company	Stiima Institute of Intelligent Industrial Techr		
Project Title	DATA MEASUREMENT SYSTEM WITH GAS SENSORS, NEW MEASUREMENT PROPOSALS FOR CONTAMINANT CONCENTRATION		
Description Main tasks	• Management of resources and personnel. • Integration of communications devices. • Development of environmental monitoring systems. • Technology for the determination of environmental pollutants.		
Sector	-Health and medical sciences. -Work and industry environmental quality. -IOT communications / data transmission.	Areas	• Systems engineering. • Support engineering. • Electronics & Embedded Software.
  			

Papers:

- Development and simulation of a low cost closed thermal system to generate thermal cycles through PID control and IOT applications, Blue Pen Journals, 01 October 2019 (In progress).
- Humidity generator system development for laboratory applications integrating a PID controller in low cost IOT applications, International Journal of Science and Research (IJSR).
- Creation of an Infrastructure of sensor networks for monitoring the environment and the person, GARR 2017 Conference Act, 2017 - Italy, ISBN/ISSN 978-88-905077-7-9, DOI 10.26314.

Results:

Prototypes:

- UCOMICASEG development of an iot station for testing and calibration of gas sensors.
- SIGECOBU Development of an IOT humidity generator system.
- CACOTE development and behavior of a PID temperature controller sensors for IOT applications.

Boards:

- Ambroom Electronic card.
- Sensors Card.
- Tension card.
- Ambilowfi Ambient Card.

Responsibilities:	• Definition of conceptual idea, development and realization • Physical and electromechanical assembly of the project prototype.
Activity	• (Documentation): Drafting of technical documents for reporting. • (Documentation): Design and concept (prototype) / industrial design. • (Electronics): Test on models and working prototype by computer simulation. • (Electronics): Control of hardware/software performance, integrated circuits. • (Electronics): Execution of PCB design tests and analysis report drafting. • (Mechanics): Creation of individual components and assembly. • (Mechanics): (structural dimensioning, kinematics, welding). • (Verification): Development of individualized test plans for each project. • (Validation): Definition and tracking of basic prototype sensor parameters.

PORTFOLIO: UCOMICASEG

Job Title on project | Duration | (11/2016) - (11/2019)

Company Biomedical technology Institute (ITB), National Research Council (CNR) (Milan, Italy).
ISTITUTO DI FISILOGIA CLINICA, ISA, IASI, ICAR, IEIT, IIT, IBFM, IAC, IBBA, IMATI.
MIUR PON ELIXIR "CNR-BIOmics", Italian flagship "InterOmics".

Project Title Development of an integrated platform for the application of "omic" sciences to biomarker definition and theranostic, predictive and diagnostic profiles.

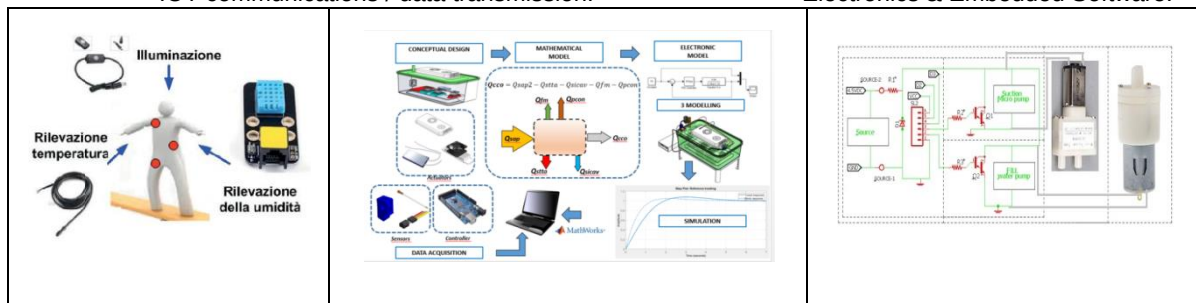
Description Main tasks

- Development of biomedical diagnostic instruments.
- Development of environmental monitoring systems.
- Technology for the determination of environmental pollutants.

Sector -Health and medical sciences.
-Work and industry environmental quality.
-IOT communications / data transmission.

Areas

- Systems engineering.
- Support engineering.
- Electronics & Embedded Software.



Papers:

-Development and simulation of a low cost closed thermal system to generate thermal cycles through PID control and IOT applications, Blue Pen Journals, 01 October 2019 (In progress).
-Humidity generator system development for laboratory applications integrating a PID controller in low cost IOT applications, International Journal of Science and Research (IJSR).
-Creation of an Infrastructure of sensor networks for monitoring the environment and the person, GARR 2017 Conference Act, 2017 - Italy, ISBN/ISSN 978-88-905077-7-9, DOI 10.26314.

Results:

Prototypes:

-UCOMICASEG development of an iot station for testing and calibration of gas sensors.
-SIGECOBU Development of an IOT humidity generator system.
-CACOTE development and behavior of a PID temperature controller sensors for IOT applications.

Boards:

- Ambroom Electronic card.
- Sensors Card.
- Tension card.
- Ambilowfi Ambient Card.

Responsibilities:

- Definition of conceptual idea, development and realization
- Physical and electromechanical assembly of the project prototype.

Activity

- (Documentation): Drafting of technical documents for reporting.
- (Documentation): Design and concept (prototype) / industrial design.
- (Electronics): Test on models and working prototype by computer simulation.
- (Electronics): Control of hardware/software performance, integrated circuits.
- (Electronics): Execution of PCB design tests and analysis report drafting.
- (Mechanics): Creation of individual components and assembly.
- (Mechanics): (structural dimensioning, kinematics, welding).
- (Verification): Development of individualized test plans for each project.
- (Validation): Definition and tracking of basic prototype sensor parameters.

Job Title on project | Duration | (04/2015) - (04/2017)



Company Colciencias; Administrative Department of Science, Technology and Innovation. University national of Colombia, physics school, Medellin, Colombia.

Project Title **INSELF generation of clean energies.**

Program "Study, analysis and development of a system of efficient generation of clean energies and the use of photovoltaic cells based on organic and inorganic cells."

**Description
Main tasks**

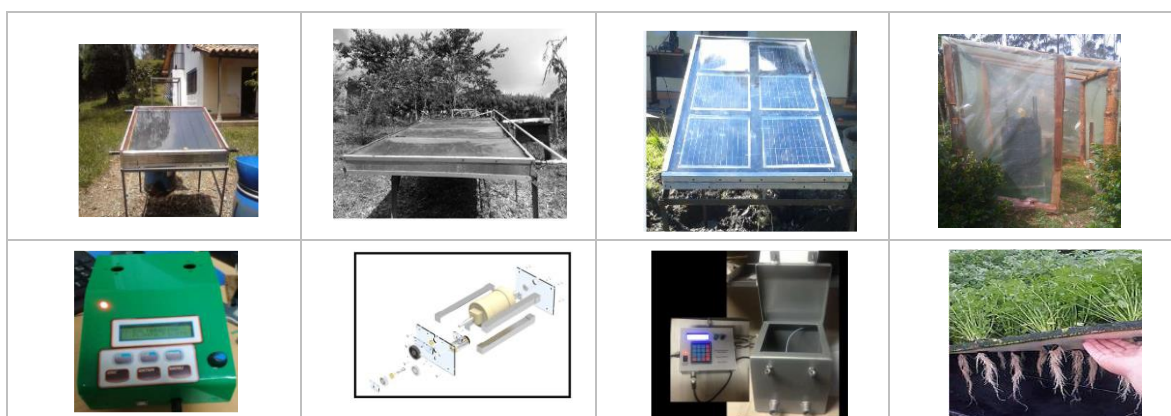
- Chemical compounds for semiconductor films, perovskites.
- Installation, maintenance and assembly of sensors in solar energy water distillers.
- Installation, maintenance and assembly of gas sensors in gas bio-digester plants.
- Installation, construction and preparation of laboratories in buildings for engineering projects, electrical installations, gas, electronics and security.

Sector

- Clean or renewable energy.
- IOT communications/ data transmission.
- Photovoltaic cell development industry.
- ew materials, thermoelectric systems.

. Areas

- Systems engineering.
- Support engineering.
- Electronics & Embedded Software.
- Photovoltaic energy.



Pilot plant:

- PISICOPUA pilot plant solar collector system.
- PISICOSO Pilot Plant Solar Distiller.
- PISIBIOME Pilot Plant Biodigester.

Results:

Prototypes:

- Prototype of measurement in the solar collector.
- Prototype Spin-coating prototype development of semiconductors films.
- Prototype for the spraying metallurgical silicon.



Responsibilities:

- Definition of conceptual idea, development and realization.
- Physical and electromechanical assembly of the project prototype.
- Construction and assembly of the proposed pilot plants.
- Training and preparation of engineering and masters students.

Activity

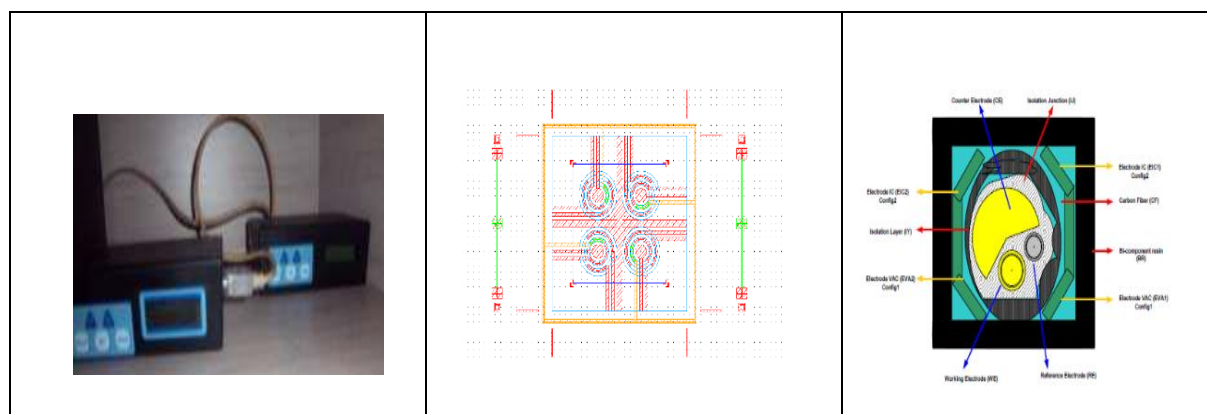
- (Electronics): Test on models and working prototype by computer simulation.
- (Verification): Development of individualized test plans for each project.
- (Electronics): Processing of product specifications with PCB models for electronic design.
- (Electronics): Development of new PCBs and related wiring of the models.
- (Mechanics): Creation of individual components and assembly.
- (Mechanics): Laminations of composite materials.
- (Mechanics): Technical Drawing (basics of CAD drawing, manual technical drawing).
- (Verification / Validation): Supervision of test team for checking sensors / devices..





Job Title on project | Duration | (09/2012) - (03/2015)

Company	Biomedical technology Institute (ITB), National Research Council (CNR) (Milan, Italy). ISTITUTO DI FISILOGIA CLINICA, ISA, IASI, ICAR, IEIT, IIT, IBFM, IAC, IBBA, IMATI.		
	MIUR PON, Industrial Research “VIRTUALAB”.		
Project Title	Advanced Biomedical Mechatronic Systems for Diagnosis and Therapy based on Virtual and Augmented Reality, microelectronics and on high throughput robotised workstations.		
Description	- Development of biomedical diagnostic instruments. 2014 “REDEMA”, “MEA construction as a support, for a cell culture plate and its measurement of activities in different areas”, 2014-2015.		
Main tasks	2013. “SAMLO”, “Construction of a wire probe with sensors to identify and eliminate neuronal tumour masses, 2012-2013.		
Sector	- Health and medical sciences. - Work and industry environmental quality. - IOT communications and data transmission.	Areas	- Systems engineering. - Support engineering. - Electronics/ Embedded Software.



Papers:

- Construction of an amperometric biosensor and instrumentation in micro electrodes (MEA) 4 wells, Polytechnic magazine, Year 9, 18 June 2013, ISSN 1900-2351.

Prototypes:

- Results:**
- Prototype instrument ARTERIO1, generator of stimuli and signals for patients with problems related to multiple atherosclerosis, IBFM, Milan Italy 2014 applications.

Boards:

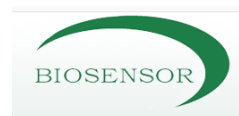
- Multimotors Card.
- Freq Sensors Card.



Responsibilities:	- Biosensors and software development. - Medical devices development.
Activity	(Documentation): Drafting of datasheet for sensors. (Documentation): RFI Initial process, to plan the negotiation, the competition or the approval of suppliers. (Documentation): RFP, criteria and characteristics, quality or safety. (Documentation): FRG / ITT Public procurement through offers or other. (Electronics): Test on models and working prototype by computer simulation. (Electronics): Electronic prototypes cards, firmware development. (Verification / Validation): Development of individualized test plans for each project.



Job Title on project | Duration | | (01/2008) - (07/2012)

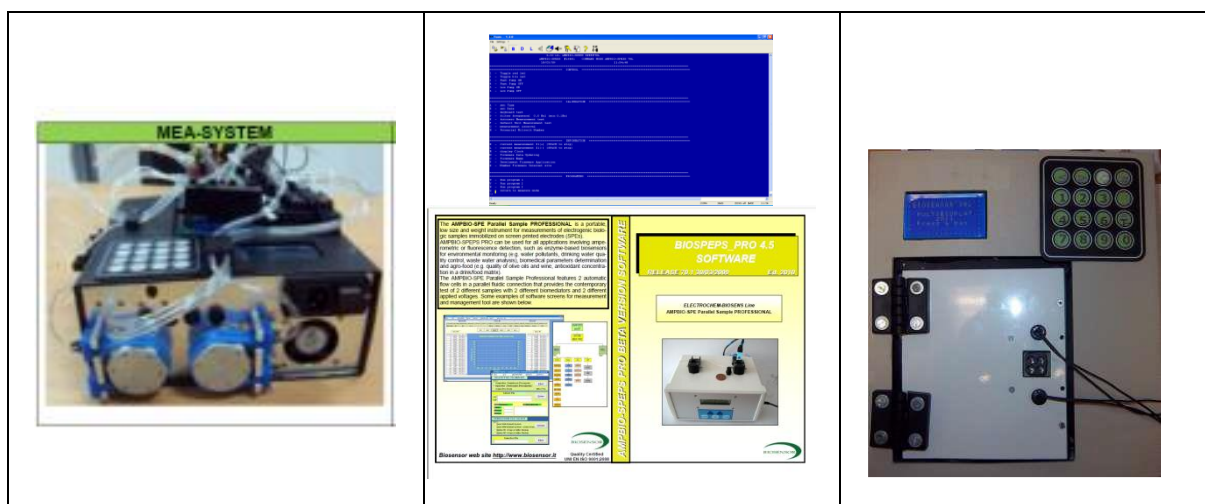


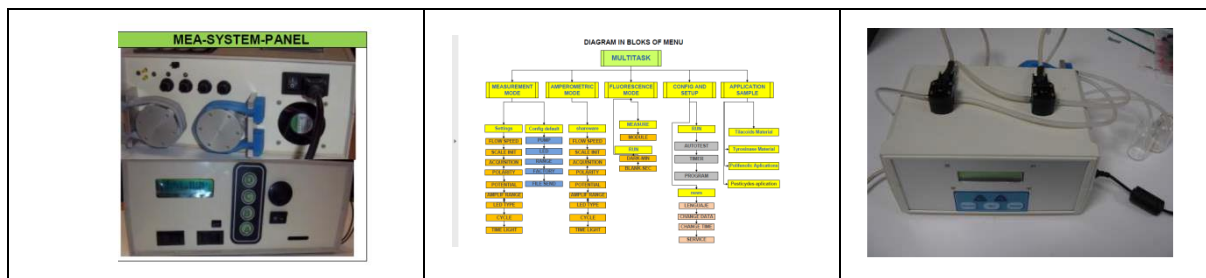
Company Biosensor Srl company, Formello- Rome, Italy.

MULTITASKS, AGROBIOSENS, BEEP-C-EN, BIOAL, NIGHT VISION, PHOTOEVOLUTION, MICROBIOSIS, BIO-NUTRA

- Project Title**
- MULTITASKS: Innovation of biosensor product in the creation of a basic system called Multitasks Biosensor and its application in Agro food.
 - AGROBIOSENS: Industrial research for the realization of biosensors for the monitoring of pollution from bio-herbicides in the agro-food sector.
 - BEEP-C-EN: Biosensor for Effective Environmental Protection and Commercialization- Enhanced, EU-FP7.
 - BIOAL: biosensor system for the determination of dangerous contaminants for human health in agro-food products, FILAS.
 - NIGHT VISION: Macular pigment extract from plant organism in nanoparticles of inorganic / organic matrix with the aim of protecting the retina optic astronauts, ASI.
 - PHOTOEVOLUTION: BIKIS Human space flight DC-VUS-2009-004 (ASI / NASA) Italian Space Agency (ASI) Mission STS-134 May 16th 2011 Center NASA.
 - MICROBIOSIS: For the realization of a completely automatic bench system, able to monitor hundreds of water samples daily, call for tenders.
 - BIO-NUTRA: Construction of an automated Photo-Bioreactor, Bando FILAS.
- Description Main tasks**
- Technical engineer multi-skilled, app: electronic and instrumentation tests.
 - Firmware development for peripheral handling.
 - Software development for process monitoring.
 - Design of electronic cards for measuring variables.

- | | | | |
|---------------|--|--------------|-----------------------------------|
| Sector | • Communications and data transmission. | Areas | • Systems engineering. |
| | • Health sciences, agro-food line. | | • Support engineering. |
| | • Work and industry environmental quality. | | • Electronics/ Embedded Software. |
| | • Computer science and information sciences. | | |
| | • Aeronautical and space projects. | | |





Papers:

- Construction of an amperometric biosensor and instrumentation in micro electrodes (MEA) 4 wells, Polytechnic magazine, Year 9, 18 June 2013, ISSN 1900-2351.
- A new embedded biosensor platform based on micro-electrodes array (MEA) technology, Sensors and Actuators B: Chemical, Volume 176, January 2013.
- Towards An integrated biosensor array for simultaneous and rapid multi-analysis of endocrine disrupting chemicals. Analytica Chimica Acta 2012.
- Biosensors based on 4-wells micro array systems: study, design, construction and applications, Sensors Lecture Notes in Electrical Engineering, Volume 162, 2014, Springer.
- "Multiparametric system for biosensing technologies applications based on microarray electrodes", Polytechnic magazine, Year 7, n° 13, 18 December 2011, ISSN 1900-2562.
- Comparison of two photosynthetic biomediators for herbicide detection, Polytechnic magazine, Year 7, n° 13, page p101-106 Medellin, 18 December 2011, ISSN 1900-2562.
- Biosensors: new frontiers for the environmental analysis, Polytechnic magazine, Year 7, n° 13, page 93- 100 Medellin, 18 December 2011, ISSN 1900-2562.
- Direct laser immobilization of photosynthetic material on screen printed electrodes for amperometric biosensor, Applied Physics Letters/Volume 98/Issue 9/BIOPHYSICS AND BIO-INSPIRED SYSTEMS, doi:10.1063/1.3562297 ,Feb 2011.
- Laser immobilization of photosynthetic material on screen printed electrodes, Lasers and Electro-Optics (CLEO), 2011 Conference on, Print ISBN: 978-1-4577-1223-4, INSPEC Accession Number:12142005, 1-6 May 2011.
- Biosensors for Effective Environmental and Agrofood Protection and Commercialization: from research to market, Microchimica Acta, SpringerLink ISSN 0026-3672, Jan 2010.

Results:

Prototypes:

- Prototype STATICMEA An instrument for the measurement of signals in fluorescence using micro electrodes 4 wells, in biological materials, project MULTIBIOPLAT, 2012, Biosensor, Rome-Italy.
- Prototype MEASYSTEM An instrument for the measurement of signals in Amperometry using micro electrodes 1 well, in biological materials, project MULTIBIOPLAT, 2011, Biosensor, Rome-Italy.

Software:

- Fluorescence and survival graphical software; MOMA (partners) 2006-2009; Product Line: FLUORESCENCE-BIOSENS; Instrument: OPTICBIO-Multicell Survival.
- Amperometric detection software, Project: SMES; Product Line: ELECTROCHEM-BIOSENS; Instrument: AMPBIO-ITOTEMP.
- Amperometric and Fluorescence detection software; Project: MULTITASK; Product Line: ELECTROCHEM-BIOSENS; Instrument: AMPBIO-ITOTEMP.

Responsibilities:

- Design, prototyping and implementation of prototypes for applications. food farming, bio-sensorial environmental.
- Development, validation, Firmware / Hardware tests.

Activity

- Documentation): Drafting of datasheet for sensors.
- (Documentation): FRG / ITT Public procurement through offers or other means.
- (Electronics): Test on models and working prototype by computer simulation.
- (Electronics): Electronic prototypes cards, firmware development.
- (Verification / Validation): Development of individualized test plans for each project.
- (Verification / Validation): Analysis of the performance of IT tools.



Job Title on project | Duration | | (01/2010) - (02/2012)

- Company**
- Crystallography Institute (IC), National Research Council (CNR) (Rome, Italy).
 - Ben-Gurion University (BGU, Israel), Centre of Bioanalysis (NIB, Romania), Nanosens(NANO, the Netherland), Biosensor (BIO, Italy), Algatechnologies (ALGATECH, Israel).

SENSBIOSYN EU project.

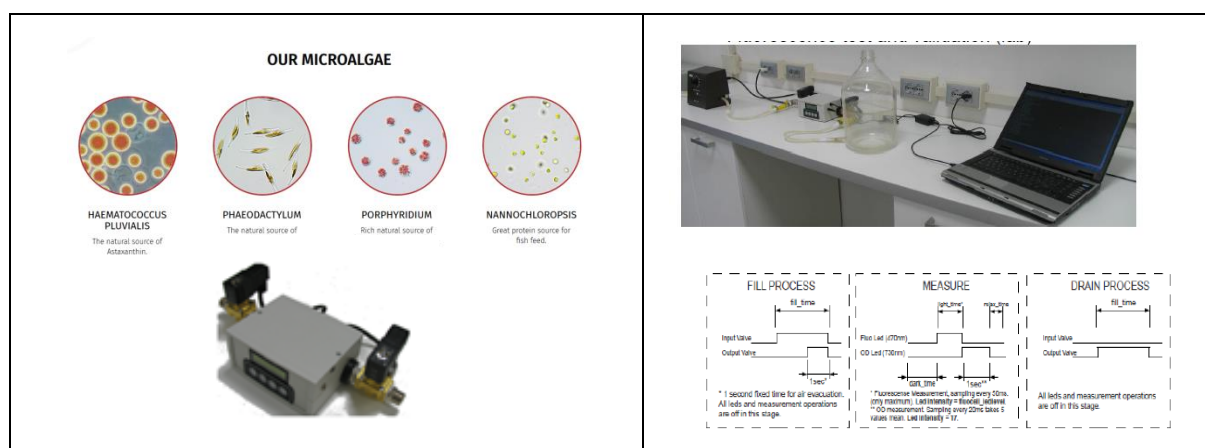
Project Title Biosensors and Sensors for the industrial biosynthesis process of widely used commercial antioxidants: nutraceuticals as additives for food and aquaculture promoting.

Description — Technical engineer multi-skilled, app: electronic and instrumentation tests.

Main tasks

- Temperature monitoring, Water sampling, Tank cleans, Disinfections.
- Firmware development for peripheral handling.

- Sector**
- Communications and data transmission.
 - Computer science and information sciences.
 - Production of algal biomass.
 - Methods for monitoring chemical species.
- Areas**
- Systems engineering.
 - Support engineering.
 - Electronics/ Embedded Software.
 - Human health and biotechnology.



Papers:

- Results:**
- Direct laser immobilization of photosynthetic material on screen printed electrodes for amperometric biosensor, Applied Physics Letters/Volume 98/Issue 9, Feb 2011.
 - Laser immobilization of photosynthetic material on screen printed electrodes, Lasers and Electro-Optics (CLEO), 2011 ISBN: 978-1-4577-1223-4.

- Responsibilities:**
- Software development for process monitoring.
 - Design of electronic cards for measuring variables.
 - Generation of reports on the assembly of flow cells and instrumentation.

- Activity**
- (Documentation): Drafting of datasheet for sensors.
 - (Documentation): RFI Initial process, to plan the negotiation, the competition or the approval of suppliers.
 - (Documentation): RFP, criteria and characteristics, quality or safety.
 - (Documentation): FRG / ITT Public procurement through offers or other.
 - (Electronics): Test on models and working prototype by computer simulation.
 - (Electronics): Electronic prototypes cards, firmware development.
 - (Verification / Validation): Development of individualized test plans for each project.



Job Title on project | Duration | | (01/2009) - (09/2009)

- Company • Crystallography Institute (IC), National Research Council (CNR) (Rome, Italy).
• Vitens (Netherlands), Nanosens(Netherlands), Biosensor(Italy), Das Italy.

Multibioplat European project.

Project Title

Innovative biosensor technology in an industrial platform that can be easily adopted for multi-parameter and multi-sensor design and production, based on electrochemical-optical transduction mechanisms, on NEMS (Nano Electro-Mechanical Systems) technology and on advanced modular techniques, to be able to offer a wide range of high performance products on the market.

Description Main tasks

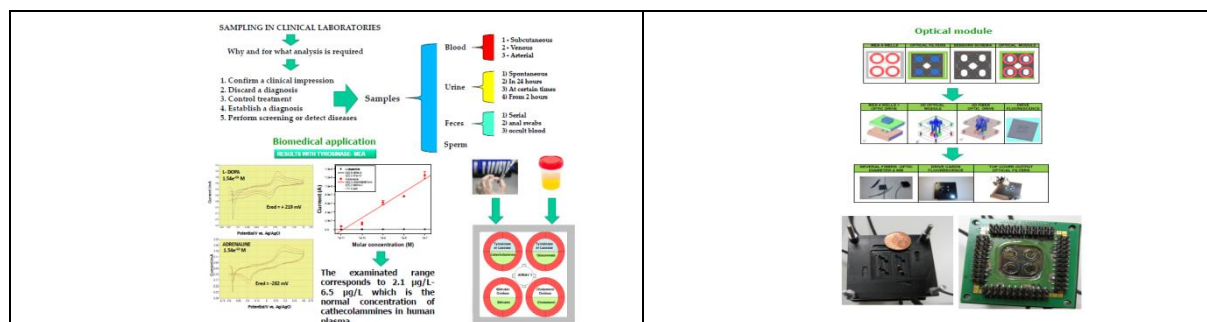
- Technical engineer multi-skilled, app: electronic and instrumentation tests.
- Development of fluorescence based optical systems.
- Use of technology such as Micro Array Electrode.
- Use of nanowires sensors to detect biological material..

Sector

- Communications and data transmission.
- Health sciences, agro-food line.
- Work and industry environmental quality.
- Computer and information sciences.

Areas

- Systems engineering.
- Support engineering.
- Electronics/ Embedded Software.
- Human health and biotechnology.



Papers:

- Biosensors based on 4-wells micro array systems: study, design, construction and applications, Sensors Lecture Notes in Electrical Engineering, Volume 162, 2014, pp 171-175, Springer.
- Multiparametric system for biosensing technologies applications based on microarray electrodes", Polytechnic magazine, Year 7, n° 13, page 107- 116 Medellin, 18 December 2011, ISSN 1900-2562.

Results:

Prototypes:

- Prototype STATICMEA An instrument for the measurement of signals in fluorescence using micro electrodes 4 wells, in biological materials, Biosensor, Rome-Italy.
- Prototype MEASYSYSTEM An instrument for the measurement of signals in Amperometry using micro electrodes 1 well, in biological materials, 2011, Biosensor, Rome-Italy.

Responsibilities:

- Software development for process monitoring.
- Design of electronic cards for measuring variables.
- Generation of reports on the assembly of flow cells and instrumentation.

Activity

- (Documentation): Drafting of datasheet for sensors.
- (Documentation): FRG / ITT Public procurement through offers or other means.
- (Electronics): Test on models and working prototype by computer simulation.
- (Electronics): Electronic prototypes cards, firmware development.
- (Verification / Validation): Development of individualized test plans for each project.
- (Verification / Validation): Analysis of the performance of IT tools.



Job Title on project | Duration | (05/2008) - (11/2008)



- Company**
- Crystallography Institute (IC), National Research Council (CNR) (Rome, Italy).
 - (MLU, Germany) of Halle–Wittenberg, (UPE, France), (UNI-PI, Italy), (NIB, Romania), (UOC, Greece), (Poland), (UNI-MI, Italy), (Enervit, Italy).

Nutra –Snacks European Project.

Project Title Ready-to-eat food for breakfast and sport activity with high content of nutraceuticals preventing disease and promoting public health.

- Description**
- Development hardware and software.
- Main tasks**
- Installation, maintenance and assembly of sensors.
 - Plant cell and in vitro culture systems together with a biotechnological, Control.

- | | | | |
|---------------|--|--------------|-----------------------------------|
| Sector | • Communications and data transmission. | Areas | • Systems engineering. |
| | • Health sciences, agro-food line. | | • Support engineering. |
| | • Computer science and information sciences. | | • Electronics/ Embedded Software. |
| | • Research. | | • Human health and biotechnology. |



Papers:

- Develop of new environmental biosensors instruments for herbicides analyses , Politecnica journal, Year 4, n° 6, January 2008, ISSN 1900-2351.
- Design of a portable instrument for environmental applications by amperometric measurements on biological material, Politecnica journal, Year 5, n° 9, December 2009 ISSN 1900-2351.

Results:

Prototypes:

- Photobioreactor Lab System.

Software

- Control system software Photobioreactor.

Responsibilities:

- Software development for process monitoring.
- Design of electronic cards for measuring variables.
- Generation of reports on the assembly of flow cells and instrumentation.

Activity

- (Documentation): Drafting of datasheet for sensors.
- (Documentation): FRG / ITT Public procurement through offers or other means.
- (Electronics): Test on models and working prototype by computer simulation.
- (Electronics): Electronic prototypes cards, firmware development.
- (Verification / Validation): Development of individualized test plans for each project.
- (Verification / Validation): Analysis of the performance of IT tools.



INDUSTRIAL WORK

04/2007 – 12/2007	REMOTE CONTROL TECHNICAL ENGINEER
<i>Project Title</i> Responsibilities	-Electrical substation ENTRERRIOS/ RIBA-ROJA/VICTORIA (Spain). - Technical engineer multi-skilled.
<i>Activity</i>	Uses / configuration PCA (Automatic configuration program), IOS, CCS, MULC. • Management system SCI (integrated control system), remote Elitel 5. • CCS configuration (substation control center) is the core of the SCI system. • IOS management (Win-XP embedded), status of all substation elements.
04/2004 – 12/2006	FIELD TECHNICAL SERVICE ENGINEER
<i>Project Title</i> Responsibilities	COPLINE new automated machines for jobs in copying companies. – Technical engineer multi-skilled, app: electronic and instrumentation tests.
<i>Activity</i>	• Technical service in photocopying machines, services: change of pinions, belts, transmission and mechanical service. Repair of electric cards. • Assembly and installation of business and home electrical networks, voltage tables of 440, 220 VAC and 110 VAC, installation of electrical panels. • PPM maintenance for thermal stamping machines.
04/2002 – 04/ 2003	TECHNICAL TELECOMMUNICATIONS ENGINEER
<i>Project Title</i> Responsibilities	EPMDYS, Last mile solutions, Bank of the Republic, Davivienda, Bank West. – Instructor, for the preparation of support group clients at departmental level.
<i>Activity</i>	• Solutions in communications and data transmission banking sector, using technology Adsl, G.hdsl, Frame Relay, X25, F.O. • Design of business IT solutions, LAN, WAN, HP OV, Cisco W, Nortel, Passport. • Computer security: TCP / IP protocols, subnets, Vpn, Dhcp, Dns, Firewall, Antivirus, Cisco Routers, 3com, SW, Avaya, HP, ISDN.
08/1997 – 05/ 2000	COMPUTER AND AUTOMATION TECHNOLOGIST
<i>Project Title</i> Responsibilities	CONBAS, injection machines orders associated with manufacturing. Firmware development for peripheral handling • Software development for process monitoring.
<i>Responsibilities</i>	Research and development of control systems and measurement of process for injection plastics industrial with software and hardware technology in microcontrollers Microchip and platform visual basic.
<i>Activity</i>	
10/1996- 07/1997	INDUSTRIAL INSTRUMENTATION TECHNOLOGIST
<i>Project Title</i>	SIRESOCOP: Solar system Refrigeration and supervision system PC, software and system controls” at the Colombian Polytechnic JIC in Colombia.
<i>Activity</i>	Technologist: The "variable plant": calibration instruments, measurement and Process Control for industrial applications.

PUBLICATIONS

1. Development and simulation of a low cost closed thermal system to generate thermal cycles through PID control and IOT applications, Blue Pen Journals, 01/10/2019.(In progress).
2. Humidity generator system development for laboratory applications integrating a PID controller in low cost IOT applications, International Journal of Science and Research (IJSR), 01/10/2019 (In progress).
3. Creation of an Infrastructure of sensor networks for monitoring the environment and the person, GARR 2017 Italy, ISBN/ISSN 978-88-905077-7-9, DOI 10.26314.
4. Analysis, studies and tests to instruments connected batteries for a long term in space, Polytechnic magazine, Year 10, Vol, 10, n° 18, Medellin, 20 June 2014.
5. Construction of an amperometric biosensor and instrumentation in micro electrodes (MEA) 4 wells, Polytechnic magazine, Year 9, n° 16, page 9- 20 Medellin, 18 June 2013, ISSN 1900-2351.
6. A new embedded biosensor platform based on micro-electrodes array (MEA) technology, Sensors and Actuators B: Chemical, Volume 176, January 2013, Pages 275–283. <http://dx.doi.org/10.1016/j.snb.2012.09.101>.
7. Towards An integrated biosensor array for simultaneous and rapid multi-analysis of endocrine disrupting chemicals. Analytica Chimica Acta.
8. Biosensors based on 4-wells micro array systems: study, design, construction and applications, Sensors Lecture Notes in Electrical Engineering, Volume 162, 2014, pp 171-175, Springer.
9. Multiparametric system for biosensing technologies applications based on microarray electrodes, Polytechnic magazine, Year 7, n° 13, page 107- 116 Medellin, 18 December 2011, ISSN 1900-2562.
10. Comparison of two photosynthetic biomediators for herbicide detection, Polytechnic magazine, Year 7, n° 13, page p101-106 Medellin, 18 December 2011, ISSN 1900-2562.
11. Biosensors: new frontiers for the environmental analysis, Polytechnic magazine, Year 7, n° 13, page 93- 100 Medellin, 18 December 2011, ISSN 1900-2562.
12. Direct laser immobilization of photosynthetic material on screen printed electrodes for amperometric biosensor, Applied Physics Letters/volume 98/issue 9/biophysics and bio-inspired systems, doi:10.1063/1.3562297 ,Feb 2011.
13. Laser immobilization of photosynthetic material on screen printed electrodes, Lasers and Electro-Optics (CLEO), 2011 Conference on, Print ISBN: 978-1-4577-1223-4, INSPEC Accession Number:12142005, 1-6 May 2011.
14. Biosensors for Effective Environmental and Agrofood Protection and Commercialization: from research to market, Microchimica Acta, SpringerLink ISSN 0026-3672, Jan 2010.
15. Design of a portable instrument for environmental applications by amperometric measurements on biological material, Polytechnic journal, Year 5, n° 9, pag 107- 114 Medellin, December 2009 ISSN 1900-2351.
16. Develop of new environmental biosensors instruments for herbicides analyses , Polytechnic journal, Year 4, n° 6, pag 116- 123, Medellin, January 2008, ISSN 1900-2351.